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标题	Recent Advances in Nickel Catalysts for Sonogashira Coupling
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摘要（原文节选）

The Sonogashira coupling, catalyzed by transition metals, has become a valuable method for forming carbon‐carbon bonds. While palladium‐copper was originally recognized as the most effective catalyst for this transformation, recent advances have highlighted the growing prominence of nickel‐catalyzed Sonogashira reactions. Nickel has gained considerable interest due to its lower cost and more environmentally friendly profile compared to palladium. This review focuses on recent progress in nickel‐catalyzed Sonogashir ⋯

数据表 · TABLE 3（10 行，含数值行 10 行）

A comparison of the Pd/NiFe2O4 catalyst and earlier catalysts used in the Sonogashira coupling between phenylacetylene and iodobenzene.

Entry	Catalyst	Solvent	T, ° C	Base	Time, h	TOF, h −1	Yield, %	Ref.
1	Pd3Cu1/SiC	DMF	60	Cs2CO3	8	—	2	[61]
2	Pd/Fe3O4@GON	DMSO	120	NaOAc	0.5	62.1a	99	[62]
3	Ni/rGO	NMP/CuI	120	K2CO3	4	—	95	[63]
4	Co‐MS@MNPs/CS	DMSO	140	KOH	10	—	72	[64]
5	Pd‐MNPSS	H2O	100	K2CO3	10	—	87	[65]
6	Pd‐Co/Graphene	THF/H2O/CuI	80	Et3N, PPh3	12	4.1a	98	[66]
7	Pd‐phosphinite polymer (0.16)	H2O	60	DABCO	24	—	89	[67]
8	Pd3Cu1/SiC under Xe lamp	DMF	60	Cs2CO3	8	8.9a	99	[61]
9	Pd/NiFe2O4	EtOH	70	K2CO3	1	106.4	99	[60]
10	Pd/NiFe2O4	H2O	70	K2CO3	24	—	85	[60]

脚注（逐字）：a TOF = mmol product × (mmol Pd)‐1 × (time)−1

正文对应句（逐字）：Wilson et al. developed Ni‐Pd bimetallic nanoparticles (NPs) supported on multi‐walled carbon nanotubes (Ni‐Pd/MWCNTs) via a solventless ball‐milling method, enabling ligand‐ and copper‐free Sonogashira reaction of Ar‐X with terminal alkynes. Optimal catalysts (e.g., 7.9 wt% Ni/0.81 wt% Pd) feature

数据表 · TABLE 6（4 行，含数值行 4 行）

A comparison of the Fe3O4@SiO2@Cyt‐Ni/Cu catalyst and earlier catalysts utilized in the Sonogashira coupling between phenylacetylene and 4‐iodotoluene.

Entry	Catalyst	Reaction condition	Yield, %	Ref.
1	Fe 3 O 4 @SiO 2 @Cyt‐Ni/Cu (0.1 mol%)	DMSO, K 2 CO 3 , 2 h, 90 ° C	93	[70]
2	PdCu@GQD@Fe3O4 (Pd 0.3 mol%, Cu 0.35 mol%)	Toluene, DABCO, 24 h, 60 ° C	99	[71]
3	MNPs@Cs‐MS‐Co (0.55 mol%)	DMSO, KOH 10 h, 140 ° C	72	[72]
4	γ‐Fe2O3@Cu(II)IL‐SB (0.4 mol%)	DMSO, K2CO3, 4 h, 90 ° C	93	[73]

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